

Nayeem Fardin

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EDUCATION

Bachelor of Science in Computer Science and Engineering
Islamic University of Technology (IUT)

Expected Sep 2026

SKILLS

Languages	Python, JavaScript, C++, Java	Databases	PostgreSQL, Redis, InfluxDB, Pinecone, MongoDB
Web & API	React, Node, Express, FastAPI	Cloud & Ops	AWS, Docker, Kubernetes, Terraform, Linux, Git
AI & ML	LangChain, PyTorch, RAG, LLMs	Sys & Tools	Kafka, RabbitMQ, Celery, Grafana, ArgoCD, Helm

EXPERIENCE

Software Engineering Intern

FastAPI, Next.js, Flutter, Postgres

Intelligent Machines

Jun 2026–Aug 2026

- Worked across 8 projects spanning retail, banking, and internal AI; sole or primary author on 4.
- Built 7 backend domains end to end in an async FastAPI/SQLAlchemy monorepo; models, migrations, repositories, services, APIs, and tests—growing coverage from 34 to 124+ tests under strict `mypy` and `ruff` gates.
- Reworked a Flutter sales-rep app as offline-first with optimistic local state, idempotent sync and reconciliation, cutting perceived latency 5x; also fixed API authZ gaps and transactional credit/payment consistency.
- Rebuilt inherited demos into multi-tenant Postgres systems with composite tenant-scoped foreign keys and RLS, moving 156 browser-side queries behind an API and eliminating IDOR and credential-exposure paths.
- Built AI systems end to end, including an operations assistant grounded on typed SQL and cited document retrieval, and a provider-agnostic vision OCR pipeline for Bangla/English NID extraction with per-field confidence scores and pluggable document extractors.

Cybersecurity Intern

Backdoor Private Limited

- Digital forensics analysis to recover sensitive deleted artifacts using **FTK Imager** and **Autopsy**.
- Security assessments on Android applications via **MobSF** and web platforms using **Burp Suite** and **SQLMap**.

PROJECTS

ShelfWatch

[GitHub](#) | [Case Study](#)

- Built a cloud-native retail shelf monitoring platform by deploying finetuned **YOLOv11** inference on **AWS EKS**.
- Scalable inference service on AWS using **Helm + ArgoCD** (GitOps-driven deployments).
- **4×** CPU inference speedup and **70-90%** cost savings via **INT8 quantization**, **AMX** and **Spot** diversification.
- Canary releases with **Argo Rollouts** + smoke tests; **Loki/Prometheus/Grafana** for logging and observability.
- DevSecOps in **GitHub Actions** with **Trivy** and **pip-audit** for secure **ECR** promotion.

PixTools

[GitHub](#) | [Case Study](#)

- Built a distributed image-processing pipeline and migrated the ingestion API to **Go (Gin)** to achieve sub-100 ms latency, writing **Celery-compatible AMQP** envelopes to **RabbitMQ** for async **Python** worker execution.
- Architected a two-tier **K3s** cluster on **AWS** via **Terraform**, integrating **Cluster Autoscaler** to dynamically provision **EC2 Spot** nodes with graceful interruption handling.
- Implemented **KEDA** based on RabbitMQ queue depth, isolating ML workloads to prevent worker starvation.
- Engineered load-shedding defenses using **Redis** idempotency and circuit breakers, validating resilience via **k6** chaos testing to gracefully absorb 9,000-request spikes and retry storms.
- Automated CI/CD with **GitHub Actions** and **OIDC**, **OpenTelemetry** trace export to **Grafana** for observability.

RetailOS Lite

[GitHub](#) | [Case Study](#)

- Architected an async inference pipeline with **FastAPI** and **BullMQ** on **Modal GPUs**, decoupling heavy vision/LLM workloads from the **Next.js** frontend.
- Built a multimodal compliance engine combining **YOLO** detection with **OpenAI** summaries, using a fraud filter (perceptual hashing, EXIF, **GPS**) to reduce inference costs.
- Implemented a **RAG** assistant using **Pinecone** vector embeddings + **PostgreSQL** context for semantic querying of visit reports grounded by deterministic logic.